

BALARAM PAI H

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PROFESSIONAL SUMMARY

Final-year CS student with strong CS fundamentals, analytical problem-solving, and full-stack engineering experience. Designed RDBMS schemas with MySQL, built production REST APIs with Flask and microservices. Proven track record across real-time systems, machine learning, and computer vision. IEEE-published researcher.

EDUCATION

BMS Institute of Technology & Management

Class of 2027

- B.E. Computer Science and Business Systems | CGPA: 9.47

Bengaluru, KA

Delhi Public School Bangalore North

Class of 2023

- Senior Secondary High School - CBSE | Grade: 93.4%

Bengaluru, KA

CERTIFICATIONS

- Programming, Data Structures and Algorithms [NPTEL] Oct 2025
- Database Management Systems [NPTEL] Apr 2025
- Theory of Cloud Computing [NPTEL] Jun 2025

SKILLS

- Languages:** Python, C++, Java, SQL, TypeScript, HTML/CSS.
- Frameworks:** Flask, React Native, FastAPI.
- Databases:** MySQL, MongoDB, FAISS.
- Tools:** Git, GitHub, Postman, Linux, VS Code, Docker.
- CS Fundamentals:** Data Structures and Algorithms, OOP, DBMS, OS, Computer Networks, REST API Design, CI/CD.
- Specializations:** Microservices Architecture, System Scalability, Real-Time Data Processing, API Design.

EXPERIENCE

Full Stack Developer Intern – IOO Labs | Nov 2025 – May 2026

- Developed **REST APIs** using **Flask** and **JWT (JSON Web Token)** authentication.
- Integrated **React Native frontend** with backend services consisting of **Microservices Architecture**.
- Optimized MongoDB** queries for efficient data retrieval.
- Collaborated using **Git-based version control** and **CI/CD** workflows following **code review** and **development best practices**.

PROJECTS

Product Inventory Management System:

- Designed and developed full-stack inventory management system with **MySQL relational database schema**, product tracking, and **stock monitoring** directly relevant to **retail commerce**.
- Implemented CRUD operations and real-time listing updates using interactive frontend workflows.
- Reduced manual **data entry time by 65%** and improved record **accuracy by 40%**.

SHIELD - Real-Time Network Threat Detection System (Ongoing):

- Developing a **Raspberry Pi-based** system for **real-time DDoS** and brute-force attack **monitoring**.
- Using the **ELK Stack (Elasticsearch, Logstash, Kibana)** for log collection, traffic analysis, and threat visualization.
- Exploring **Isolation Forest** for anomaly detection in suspicious network activity

AI-Based Dropout Prediction & Counseling System:

- Built an end-to-end **web-based** system to identify at-risk students using **predictive analytics** and **machine learning models**.
- Integrated **XG-Boost ML model (93.4% accuracy)** into a Flask-based backend
- Developed dashboard for administrators to **monitor and trigger automated** counseling workflows.

OpenCV Gesture Controls:

- Developed real-time **computer vision** applications using **OpenCV** and **MediaPipe**.
- Built gesture-based systems for volume control, finger counting, posture tracking, and screenshot automation.
- Enabled **robust gesture recognition** and **posture tracking**, achieving **95%+** gesture recognition **accuracy at 30 FPS**.

PUBLICATION

AI-Based Dropout Prediction and Counselling System

- 10th IEEE International Conference on Smart Structures and Systems (ICSSS 2025) – 2025

LEADERSHIP

Tech Head — IEEE AESS, BMSIT&M

- Leading development of a **3U CubeSat** for radiation sensing as **Payload Co-Lead** across a **5-member team**.